

Parametric Hypothesis Tests — Part 4

Difference of Means and Large-Sample Tests

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3.4 — Tests for the Difference of Means

Setup

We have **two independent** Normal populations:

$$X_1 \sim N(\mu_1, \sigma_1) \quad X_2 \sim N(\mu_2, \sigma_2)$$

Two independent random samples:

- From population 1: n_1 observations $\rightarrow \bar{X}_1, S_1'^2$
- From population 2: n_2 observations $\rightarrow \bar{X}_2, S_2'^2$

Goal: Test hypotheses about $\mu_1 - \mu_2$ (usually testing if $\mu_1 - \mu_2 = 0$, i.e., if the two means are equal).

The General Hypotheses

H_0	H_1	Type
$\mu_1 - \mu_2 = d_0$	$\mu_1 - \mu_2 \neq d_0$	Two-tailed
$\mu_1 - \mu_2 \leq d_0$	$\mu_1 - \mu_2 > d_0$	Right-tailed
$\mu_1 - \mu_2 \geq d_0$	$\mu_1 - \mu_2 < d_0$	Left-tailed

Often $d_0 = 0$, i.e., we test whether the two population means are equal.

Case A: Both σ_1 and σ_2 Known

Test for $\mu_1 - \mu_2$ — variances known

Test statistic:

$$Z_0 = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \underset{\text{under } H_0}{\sim} N(0, 1)$$

Critical regions use z_α and $z_{\alpha/2}$ from the Normal table, as in the one-sample case.

Case B: Variances Unknown but Equal ($\sigma_1^2 = \sigma_2^2$)

Test for $\mu_1 - \mu_2$ — variances unknown, assumed equal

We estimate the **common variance** using the **pooled estimator**:

$$S_p'^2 = \frac{(n_1 - 1)S_1'^2 + (n_2 - 1)S_2'^2}{n_1 + n_2 - 2}$$

Test statistic:

$$T_0 = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S_p' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \underset{\text{under } H_0}{\sim} t_{(n_1+n_2-2)}$$

The degrees of freedom are $\nu = n_1 + n_2 - 2$. Critical values from Table 7.

When Can We Assume Equal Variances?

In practice, the assumption $\sigma_1^2 = \sigma_2^2$ can be tested using an *F*-test (not in our syllabus), or it may be given in the problem statement.

Practical rule: If the ratio of the larger sample variance to the smaller is less than about 2–3, the equal-variance assumption is often reasonable.

When the problem says “assume equal variances” or “assume $\sigma_1 = \sigma_2$ ”, use the pooled estimator.

Example 4: Difference of Means (Equal Variances)

Two production lines manufacture the same component. To compare average lengths, independent random samples are taken:

- **Line 1:** $n_1 = 12$, $\bar{x}_1 = 50.3$ mm, $s'_1 = 1.2$ mm
- **Line 2:** $n_2 = 15$, $\bar{x}_2 = 49.7$ mm, $s'_2 = 1.4$ mm

Assuming equal population variances and Normal populations, test whether the mean lengths differ at $\alpha = 0.05$.

Example 4: Solution — Step 1

Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_1 : \mu_1 - \mu_2 \neq 0$ (two-tailed)

Pooled variance:

$$s_p'^2 = \frac{(12 - 1)(1.2)^2 + (15 - 1)(1.4)^2}{12 + 15 - 2} = \frac{11 \times 1.44 + 14 \times 1.96}{25}$$

$$= \frac{15.84 + 27.44}{25} = \frac{43.28}{25} = 1.7312$$

$$s_p' = \sqrt{1.7312} \approx 1.3157$$

Example 4: Solution — Step 2

Test statistic: $T_0 \sim t_{(25)}$

$$\begin{aligned}
 t_{obs} &= \frac{(50.3 - 49.7) - 0}{1.3157 \sqrt{\frac{1}{12} + \frac{1}{15}}} = \frac{0.6}{1.3157 \times \sqrt{0.0833 + 0.0667}} \\
 &= \frac{0.6}{1.3157 \times \sqrt{0.15}} = \frac{0.6}{1.3157 \times 0.3873} = \frac{0.6}{0.5095} \approx 1.18
 \end{aligned}$$

Example 4: Solution — Step 3

Rejection region (two-tailed, $\alpha = 0.05$, $\nu = 25$):

From Table 7: $t_{0.025,(25)} = 2.060$

Reject if $|t_{obs}| \geq 2.060$.

Since $|1.18| = 1.18 < 2.060$, we **do not reject** H_0 .

Conclusion: At 5%, there is **no evidence** of a difference in mean lengths between the two production lines.

Case C: Variances Unknown, Large Samples

Test for $\mu_1 - \mu_2$ — variances unknown, $n_1 > 30$ and $n_2 > 30$

When both samples are large, the CLT applies and we can use:

$$Z_0 = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{S_1'^2}{n_1} + \frac{S_2'^2}{n_2}}} \underset{\text{under } H_0}{\overset{\sim}{\sim}} N(0, 1)$$

No assumption about equal variances is needed.

Example 5: Difference of Means (Large Samples)

A company wants to compare the average monthly sales (in thousands of euros) of two regions:

- **Region A:** $n_1 = 50$, $\bar{x}_1 = 87.3$, $s'_1 = 12.5$
- **Region B:** $n_2 = 45$, $\bar{x}_2 = 82.1$, $s'_2 = 14.8$

Test $H_0 : \mu_1 \leq \mu_2$ vs. $H_1 : \mu_1 > \mu_2$ at $\alpha = 0.05$.

Example 5: Solution

Hypotheses: $H_0 : \mu_1 - \mu_2 \leq 0$ vs. $H_1 : \mu_1 - \mu_2 > 0$ (right-tailed)

$$\begin{aligned}
 z_{obs} &= \frac{(87.3 - 82.1) - 0}{\sqrt{\frac{(12.5)^2}{50} + \frac{(14.8)^2}{45}}} = \frac{5.2}{\sqrt{\frac{156.25}{50} + \frac{219.04}{45}}} \\
 &= \frac{5.2}{\sqrt{3.125 + 4.868}} = \frac{5.2}{\sqrt{7.993}} = \frac{5.2}{2.827} \approx 1.84
 \end{aligned}$$

Rejection region (right-tailed, $\alpha = 0.05$): $z_{obs} \geq z_{0.05} = 1.645$

Since $1.84 > 1.645$, we **reject** H_0 .

p-value:

$$p = P(Z > 1.84) = 1 - \Phi(1.84) = 1 - 0.9671 = 0.0329$$

Summary: Tests for $\mu_1 - \mu_2$

Case	Conditions	Test Statistic	Distribution
A	σ_1, σ_2 known	Z_0 with σ_1, σ_2	$N(0, 1)$
B	σ_1, σ_2 unknown, equal	T_0 with S'_p	$t_{(n_1+n_2-2)}$
C	Large samples ($n_1, n_2 > 30$)	Z_0 with S'_1, S'_2	$\dot{\sim} N(0, 1)$

3.5 — Large-Sample Tests (Asymptotic Normality)

Motivation

So far, we assumed **Normal populations** (except for Case 2' and Case C, where the CLT helped).

What if we do **not know** the population distribution, but have a **large sample** ($n > 30$)?

By the **Central Limit Theorem**, the sampling distribution of $\bar{X}$ is approximately Normal, regardless of the population distribution.

This allows us to perform tests even without the Normality assumption.

Test for μ — Large Sample, Non-Normal Population

Test for μ — large sample, any distribution

If $n > 30$, by the CLT:

$$Z_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}} \underset{\text{under } H_0}{\overset{\sim}{\sim}} N(0, 1)$$

This is the **same formula** as Case 2' (Normal population, σ unknown, $n > 30$), but now the Normality assumption on the population is relaxed.

The critical regions and p -value calculations use the Standard Normal table.

Example 6: Large-Sample Test for μ

A logistics company records the delivery time for packages. The distribution is unknown (likely right-skewed). A random sample of $n = 100$ deliveries gives $\bar{x} = 3.2$ days and $s' = 0.8$ days.

Test $H_0 : \mu \leq 3$ vs. $H_1 : \mu > 3$ at $\alpha = 0.05$.

Example 6: Solution

Even though the population is not Normal, $n = 100 > 30$, so the CLT applies.

$$z_{obs} = \frac{3.2 - 3}{0.8 / \sqrt{100}} = \frac{0.2}{0.08} = 2.50$$

Rejection region (right-tailed, $\alpha = 0.05$): $z_{obs} \geq 1.645$

Since $2.50 > 1.645$, we **reject** H_0 .

***p*-value:**

$$p = P(Z > 2.50) = 1 - \Phi(2.50) = 1 - 0.9938 = 0.0062$$

Strong evidence that the mean delivery time exceeds 3 days.

Test for a Proportion p — Bernoulli Population

Test for p — large sample from a Bernoulli population

Let $X \sim \text{Bernoulli}(p)$. The sample proportion is $\hat{p} = X/n$, where $X = \sum_{i=1}^n X_i$.

For large n (typically $np_0 \geq 5$ and $n(1 - p_0) \geq 5$), by the CLT:

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} \underset{\text{under } H_0}{\overset{\sim}{\sim}} N(0, 1)$$

Note: Under H_0 , we use p_0 (not $\hat{p}$) in the denominator, because we assume H_0 is true when computing the test statistic.

Example 7: Test for a Proportion

A company claims that at most 5% of its products are defective. In a random sample of $n = 200$ items, 15 are found to be defective.

Test the company's claim at $\alpha = 0.05$.

Example 7: Solution

Hypotheses: $H_0 : p \leq 0.05$ vs. $H_1 : p > 0.05$ (right-tailed)

$$\hat{p} = 15/200 = 0.075, p_0 = 0.05$$

Checking conditions: $np_0 = 200 \times 0.05 = 10 \geq 5 \checkmark$ and $n(1 - p_0) = 190 \geq 5 \checkmark$

$$\begin{aligned} z_{obs} &= \frac{0.075 - 0.05}{\sqrt{\frac{0.05 \times 0.95}{200}}} = \frac{0.025}{\sqrt{\frac{0.0475}{200}}} = \frac{0.025}{\sqrt{0.0002375}} \\ &= \frac{0.025}{0.01541} \approx 1.62 \end{aligned}$$

Example 7: Solution (cont.)

Rejection region (right-tailed, $\alpha = 0.05$): $z_{obs} \geq 1.645$

Since $1.62 < 1.645$, we **do not reject** H_0 .

p -value:

$$p = P(Z > 1.62) = 1 - \Phi(1.62) = 1 - 0.9474 = 0.0526$$

Conclusion: At 5%, there is **marginal** but insufficient evidence to reject the company's claim. The p -value (0.0526) is just barely above α .

This is a “borderline” case. At $\alpha = 0.10$ we would reject H_0 .

Example 8: Test for Proportion (Two-tailed)

In a survey of $n = 400$ voters, 224 favor a certain policy. Test whether the true proportion differs from 50% at $\alpha = 0.01$.

Example 8: Solution

$H_0 : p = 0.50$ vs. $H_1 : p \neq 0.50$ (two-tailed)

$$\hat{p} = 224/400 = 0.56$$

$$z_{obs} = \frac{0.56 - 0.50}{\sqrt{\frac{0.50 \times 0.50}{400}}} = \frac{0.06}{\sqrt{0.000625}} = \frac{0.06}{0.025} = 2.40$$

Rejection region (two-tailed, $\alpha = 0.01$): $|z_{obs}| \geq z_{0.005} = 2.576$

Since $|2.40| = 2.40 < 2.576$, we **do not reject** H_0 at 1%.

p-value: $p = 2 \times P(Z > 2.40) = 2 \times 0.0082 = 0.0164$

At $\alpha = 0.05$ we **would** reject. At $\alpha = 0.01$ we **do not**.

Comprehensive Summary

All Test Cases — Complete Reference

#	Parameter	Conditions	Test Statistic	Distribution
1	μ	Normal, σ known	$\frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	$N(0, 1)$
2	μ	Normal, σ unknown	$\frac{\bar{X} - \mu_0}{S' / \sqrt{n}}$	$t_{(n-1)}$
2'	μ	Normal, σ unknown, $n > 30$	$\frac{\bar{X} - \mu_0}{S' / \sqrt{n}}$	$\dot{\sim} N(0, 1)$
3	σ^2	Normal	$\frac{(n-1)S'^2}{\sigma_0^2}$	$\chi_{(n-1)}^2$
4a	$\mu_1 - \mu_2$	Normal, σ_1, σ_2 known	$\frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$	$N(0, 1)$
4b	$\mu_1 - \mu_2$	Normal, $\sigma_1 = \sigma_2$ unknown	$\frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S'_p \sqrt{1/n_1 + 1/n_2}}$	$t_{(n_1+n_2-2)}$
4c	$\mu_1 - \mu_2$	Large samples	$\frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{S_1'^2/n_1 + S_2'^2/n_2}}$	$\dot{\sim} N(0, 1)$
5	μ	Any distrib., $n > 30$	$\frac{\bar{X} - \mu_0}{S' / \sqrt{n}}$	$\dot{\sim} N(0, 1)$

Statistics II — Parametric Hypothesis Tests

Critical Regions — Quick Reference

For a test statistic T with observed value t_{obs} :

H_1	Rejection region	p -value
$\theta \neq \theta_0$	$\ t_{obs}\ \geq c_{\alpha/2}$	$2 \times P(T \geq \ t_{obs}\)$
$\theta > \theta_0$	$t_{obs} \geq c_{\alpha}$	$P(T \geq t_{obs})$
$\theta < \theta_0$	$t_{obs} \leq -c_{\alpha}$ (or $\leq c_{1-\alpha}$ for χ^2)	$P(T \leq t_{obs})$

c_{α} is the critical value at level α from the appropriate table (z_{α} , $t_{\alpha,(\nu)}$, or $q_{\alpha,(\nu)}$).

The General Procedure — Revisited

Step-by-step (for any case)

1. **Identify** the parameter, population(s), and what is known/unknown
2. **State** H_0 and H_1
3. **Choose** α
4. **Select** the appropriate test statistic (from the summary table)
5. **Determine** the critical region (using the correct table and type of test)
6. **Compute** t_{obs} (and the p -value, if requested)
7. **Decide**: reject or do not reject H_0
8. **Conclude** in context

Additional Practice

Exercise C

In a random sample of $n = 250$ consumers surveyed, 75 stated they prefer brand A.

- a. Test $H_0 : p = 0.35$ vs. $H_1 : p \neq 0.35$ at $\alpha = 0.05$.
- b. Compute the p -value.

Exercise C: Solution

$$\hat{p} = 75/250 = 0.30, p_0 = 0.35$$

$$z_{obs} = \frac{0.30 - 0.35}{\sqrt{\frac{0.35 \times 0.65}{250}}} = \frac{-0.05}{\sqrt{0.00091}} = \frac{-0.05}{0.03017} \approx -1.66$$

Rejection region (two-tailed, $\alpha = 0.05$): $|z_{obs}| \geq 1.96$

Since $|-1.66| = 1.66 < 1.96$, we **do not reject** H_0 .

***p*-value:**

$$p = 2 \times P(Z \geq 1.66) = 2 \times (1 - 0.9515) = 2 \times 0.0485 = 0.0970$$

No evidence at 5% that $p \neq 0.35$.

Exercise D

Two factories produce steel rods. Quality control takes samples:

- **Factory 1:** $n_1 = 10$, $\bar{x}_1 = 502.1$ mm, $s'_1 = 4.5$ mm
- **Factory 2:** $n_2 = 12$, $\bar{x}_2 = 498.3$ mm, $s'_2 = 5.1$ mm

Assume Normal populations with equal variances. Test whether Factory 1 produces longer rods at $\alpha = 0.05$.

Exercise D: Solution

$H_0 : \mu_1 - \mu_2 \leq 0$ vs. $H_1 : \mu_1 - \mu_2 > 0$ (right-tailed)

Pooled variance:

$$s_p'^2 = \frac{9 \times (4.5)^2 + 11 \times (5.1)^2}{10 + 12 - 2} = \frac{9 \times 20.25 + 11 \times 26.01}{20} = \frac{182.25 + 286}{20}$$

$$s_p' = \sqrt{23.418} \approx 4.839$$

$$t_{obs} = \frac{(502.1 - 498.3) - 0}{4.839 \sqrt{\frac{1}{10} + \frac{1}{12}}} = \frac{3.8}{4.839 \times \sqrt{0.1833}} = \frac{3.8}{4.839 \times 0.4282} = \frac{3.8}{2.07}$$

Exercise D: Solution (cont.)

Rejection region (right-tailed, $\alpha = 0.05$, $\nu = 20$):

From Table 7: $t_{0.05,(20)} = 1.725$

Since $1.83 > 1.725$, we **reject** H_0 .

p -value: From Table 7, $t_{0.05,(20)} = 1.725$ and $t_{0.025,(20)} = 2.086$.

Since $1.725 < 1.83 < 2.086$: $0.025 < p < 0.05$.

Conclusion: At 5%, there is evidence that Factory 1 produces longer rods on average than Factory 2.

Summary of Today's Lecture

Key takeaways — Lecture 7

- **Section 3.3:** Tests for μ (σ known/unknown) and σ^2 of Normal populations use the same pivot variables as confidence intervals (Z , T , Q)
- **Section 3.4:** Tests for $\mu_1 - \mu_2$ require identifying whether variances are known, unknown-but-equal (pooled), or whether samples are large
- **Section 3.5:** Large-sample tests ($n > 30$) work for any distribution thanks to the CLT; tests for proportions use $\hat{p}$ and p_0
- In **all cases**, the procedure is the same 8-step process
- The p -value provides the strength of evidence and can be used alongside (or instead of) the critical region approach

What's Next

The second midterm covers Topics 2.2, 3, and 4.

Next: **Topic 4 — The Linear Regression Model.**

Disclaimer

These slides are a free adaptation of the course material for Estatística II by Prof. Teresa Ferreira and Prof. Sandra Custódio from the Lisbon Accounting and Business School — Polytechnic University of Lisbon.

Primary reference: Newbold, P., Carlson, W. & Thorne, B. — *Statistics for Business and Economics*, Global Edition.